

Substrate Hall-algebra engine v0.1 — Green coproduct + the physical process-dictionary (multiplication=fusion, coproduct=decay, grading=conservation)

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Engine v0.1 — Green coproduct + the process-dictionary

0. The unifying insight, made precise

Keeper's synthesis: a Ringel-Hall algebra is an algebra of processes — its multiplication counts module extensions, so the product IS interaction. This doc makes that precise by writing down the full bialgebra (multiplication AND coproduct) for the substrate Hall algebra $H(Q_{B2})$ over $GF(2)$, and reading off the physical dictionary. This is engine step 1 of 4 (coproduct $\rightarrow$ canonical basis $\rightarrow$ R-matrix $\rightarrow$ negative part).

The bet (Keeper's discipline note): "extensions = SM vertices, affine tubes = generations" is well-motivated standard math, but the PHYSICS identification must be earned. Falsification points: E0 (Elie — does finite B_2 reproduce Serre constants?) and 3-tubes = 3 generations. This doc develops the structure; the physics claims are tier-flagged and earned downstream.

1. The Ringel-Hall bialgebra (Green's theorem)

For a hereditary algebra (path algebra of acyclic quiver Q) over F_q , the Hall algebra $H(Q)$ is a twisted bialgebra:

1.1 Multiplication (fusion)

$$u_M \cdot u_L = q^{\langle \dim M, \dim L \rangle} \sum_X g^X_{ML} u_X$$

where $g^X_{ML} = \# \{ \text{submodules } N \subseteq X : N \cong L \text{ and } X/N \cong M \}$ (Hall numbers), and $\langle \cdot, \cdot \rangle$ is the Euler form $\langle \alpha, \beta \rangle = \sum_i \alpha_i \beta_i - \sum_{\text{arrows } i \rightarrow j} \alpha_i \beta_j$.

Physical reading: $u_M \cdot u_L \rightarrow u_X$ is a FUSION vertex — two incoming representations (particles) M, L combine into an extension X . The Hall number g^X_{ML} is the vertex multiplicity (how many ways M, L fuse into X); the q -power is the coupling weight.

1.2 Comultiplication (decay/emission) — the step-1 deliverable

Green's coproduct:

$$\Delta(u_X) = \sum_{M,L} q^{\langle \dim M, \dim L \rangle} (a_M a_L / a_X) g^X_{ML} \cdot u_M \otimes u_L$$

where $a_M = |\text{Aut}(M)|$ (automorphism count). On the substrate field $GF(2)$, a_M is a product of $|GL_k(F_2)|$ factors over the indecomposable multiplicities.

Physical reading: $\Delta(u_X) = \sum u_M \otimes u_L$ is a DECAY/EMISSION vertex — one incoming X splits into outgoing M, L . The same Hall number g^X_{ML} appears (crossing symmetry — fusion and decay share the vertex), reweighted by automorphism factors (the symmetry factors of the in/out states). This is the substrate's emission amplitude.

1.3 Bialgebra compatibility (Green) = crossing consistency

Green's theorem: Δ is an algebra homomorphism (twisted) — $\Delta(u_M \cdot u_L) = \Delta(u_M) \cdot \Delta(u_L)$. Physically this is crossing symmetry / unitarity consistency: the fusion of decays equals the decay of fusions. It's the substrate-level statement that the vertex is consistent under particle exchange in/out.

2. The grading = conservation laws

$H(Q)$ is graded by the dimension vector: $\deg(u_M) = \dim M \in \mathbb{Z}^{\{Q_0\}_{\geq 0}}$. Multiplication and comultiplication respect the grading: $u_M \cdot u_L$ has degree $\dim M + \dim L$ (additive) - $\Delta(u_X)$ splits degree: only (M, L) with $\dim M + \dim L = \dim X$ appear

Physical reading: the dimension vector is CONSERVED in every process. For Q_{B2} the dimension vector lives in $\mathbb{Z}^2$ (the two vertices = the two Cartan factors = S^1 Pin(2) phase + S^4 SO(5) spatial). So the two conserved “charges” carried by the dimension vector are the substrate analogues of the two Cartan quantum numbers. This is the substrate's conservation-law structure: the grading IS the conserved quantum numbers. (Which physical charges these are — relation to σ_{BF} charge, weak isospin — is the $L2 \leftrightarrow L1$ bridge, tier-flagged FRAMEWORK pending the $A_{sub} \leftrightarrow H(Q_{B2})$ map.)

3. Indecomposables = elementary; decomposables = composite

- Indecomposable modules ($\dim$ vector = a positive root) = ELEMENTARY excitations. For finite B_2 : 4 positive roots $\rightarrow$ 4 indecomposables $\rightarrow$ the elementary list is too small for the full SM (this is exactly the tell, per Keeper E0, that we need the AFFINE $\hat{B}_2$, whose tubes give infinitely many indecomposables = the generation tower).
- Decomposable modules ($M = \oplus$ indecomposables) = MULTI-PARTICLE / COMPOSITE states. The co-product Δ resolves a composite into its constituents = the substrate's compositeness statement (hadrons as bulk-K-type composites get their decay structure here).

Why affine (E0's tell, made precise): finite B_2 has 4 indecomposables — enough to reproduce the Serre constants (Elie E0 verifies this) but NOT enough for a generation tower. The affine $\hat{B}_2$ has tubes (P^1 -families of indecomposables); the NUMBER OF TUBES is the candidate generation count. If #tubes = 3, that FORCES generations (the deepest open gate). This is the bet's sharpest test.

4. The process-dictionary (engine summary)

Hall-algebra structure	Physical process	Substrate role
multiplication $u_M \cdot u_L = \sum g^X_{\{ML\}} u_X$	FUSION vertex ($M+L \rightarrow X$)	interaction amplitude
comultiplication $\Delta(u_X) = \sum u_M \otimes u_L$	DECAY/EMISSION vertex ($X \rightarrow M+L$)	decay amplitude
Hall number $g^X_{\{ML\}}$	vertex multiplicity	coupling strength (combinatorial)
Euler-form twist $q^{\langle, \rangle}$	coupling weight	the $q=2$ (field-size) weighting
grading by $\dim$ vector	conservation laws	conserved Cartan quantum numbers
bialgebra compatibility (Green)	crossing symmetry / unitarity	vertex consistency
indecomposables (positive roots)	elementary particles	the 4 finite + affine tower
affine tubes	generation tower	candidate: #tubes = 3 generations
R-matrix (engine step 3)	scattering S-matrix	exchange/braiding [next]
canonical basis (engine step 2)	the natural particle basis	derives the 5-tuples [next]

5. What this unlocks (and what it doesn't yet)

Unlocks (structural): the SM vertices are now LOCATED — they are the Hall structure constants (fusion) and the Green coproduct (decay). Goal 2 (model the SM process) is, structurally, “compute these for the substrate quiver.” The dynamics layer L3 has a precise home.

Does NOT yet deliver: - The EXPLICIT structure constants beyond the Serre relations (the full PBW table — needs Elie E0 + the affine extension). - The canonical basis (step 2) — which would DERIVE the particle basis rather than hand-assign it. - The R-matrix (step 3) — the actual scattering S-matrix. - The physical identification of the grading charges with σ_{BF} /isospin (the $A_{sub} \leftrightarrow H$ bridge).

6. Honest scope + tier

RIGOROUS (standard math): Green's theorem (Hall algebra is a twisted bialgebra); multiplication + coproduct formulas; grading by dim vector; indecomposable=root correspondence; the $q^{\langle, \rangle}$ twist.

FRAMEWORK (the bet — physical identification): multiplication=fusion-vertex, coproduct=decay-vertex, grading=conservation-laws, affine-tubes=generations. These are well-motivated (a Hall algebra genuinely is an algebra of extensions = processes) but the PHYSICS identification is earned downstream at E0 (Serre reproduction — Elie) and the 3-tubes test. Tier-flagged per Keeper's discipline note. NOT claiming the SM dynamics are derived — claiming the engine's structure is in place and its physical reading is the testable bet.

Cal #27 / honesty: this is at peak-elegance (the three goals collapse to one object). The discipline: the elegance is the bet, not the proof. E0 and 3-tubes are where it's won or lost; a clean break is valuable.

Next (my lane): engine step 2 — the canonical/crystal basis of $U_{q^+}(B_2)$ (the natural particle basis; the missed-opportunity item that would turn the 5-tuple from labeling into derivation). Then step 3 R-matrix (scattering), step 4 the negative part (antiparticles).

Cross-CI: feeds Elie E0 (the multiplication this doc formalizes is what E0 computes explicitly) + Grace Periodic Table (the object-list $\times$ multiplication table) + the derived-vs-assigned flag (canonical basis, step 2, is what flips cells from assigned to derived).

— Lyra, Engine v0.1 Green coproduct + process-dictionary filed (Phase I step 1/4). The Ringel-Hall bialgebra made precise: multiplication = fusion vertex, Green coproduct = decay/emission vertex, dim-vector grading = conservation laws, indecomposables = elementary particles, affine tubes = generation-tower candidate. SM vertices now LOCATED as Hall structure constants — Goal 2 has a precise home. The physical identification is the bet (earned at E0 + 3-tubes). Next: canonical basis (step 2) to derive the particle basis.